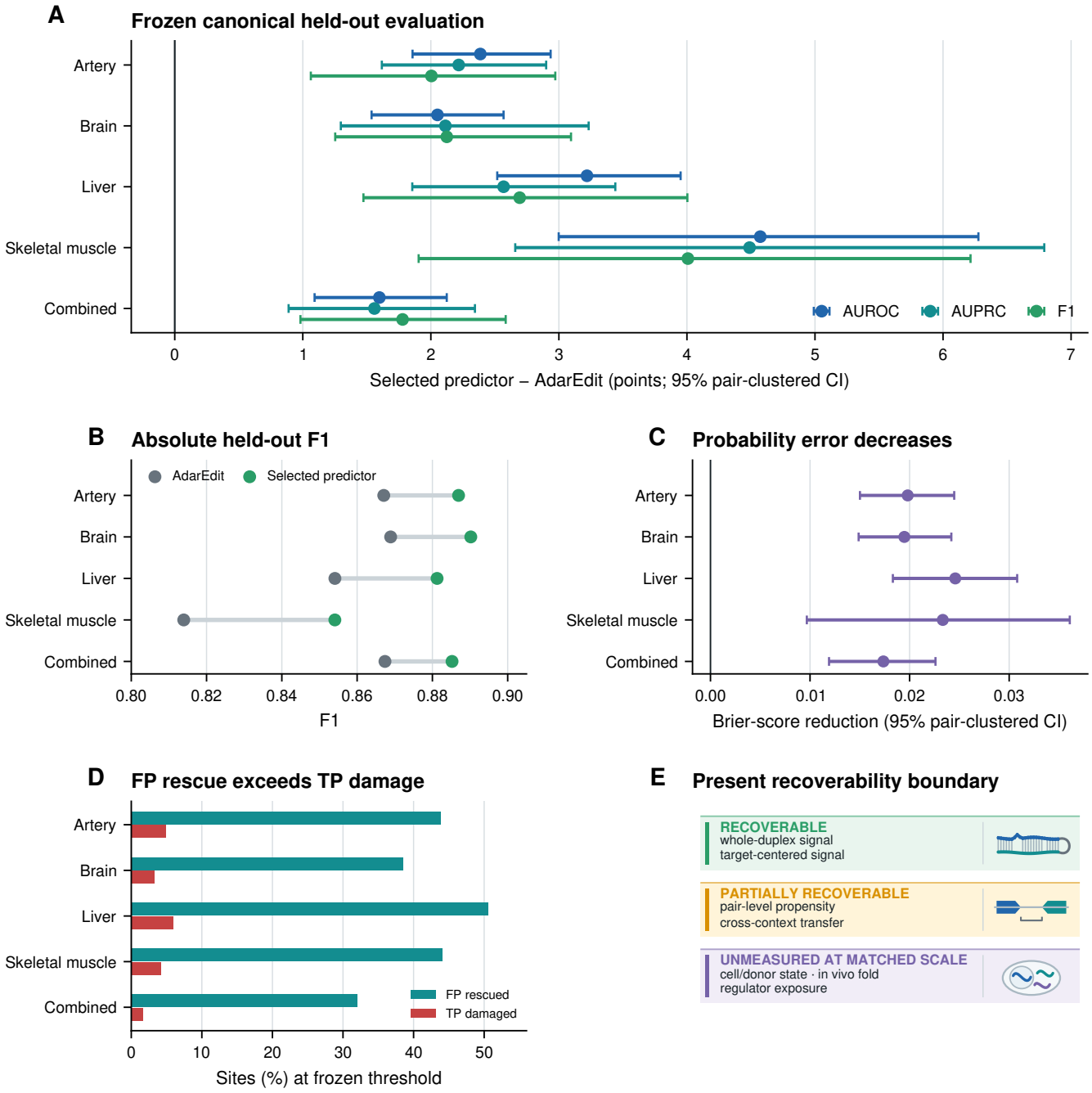
A
Frozen canonical held-out evaluation
Artery
Brain
Liver
Skeletal muscle
Combined
AUROC
AUPRC
F1
0
1
2
3
4
5
6
7
Selected predictor – AdarEdit (points; 95% pair-clustered CI)
B
Absolute held-out F1
AdarEdit
Selected predictor
Artery
Brain
Liver
Skeletal muscle
Combined
0.80
0.82
0.84
0.86
0.88
0.90
F1
C
Probability error decreases
Artery
Brain
Liver
Skeletal muscle
Combined
0.00
0.01
0.02
0.03
Brier-score reduction (95% pair-clustered CI)
D
FP rescue exceeds TP damage
Artery
Brain
Liver
Skeletal muscle
Combined
FP rescued
TP damaged
0
10
20
30
40
50
Sites (%) at frozen threshold
E
Present recoverability boundary
RECOVERABLE
whole-duplex signal
target-centered signal
PARTIALLY RECOVERABLE
pair-level propensity
cross-context transfer
UNMEASURED AT MATCHED SCALE
cell/donor state · in vivo fold
regulator exposure